

KEITH GEOFFREY NG

SENIOR AI/AGENTIC ENGINEER & CLOUD NATIVE FULLSTACK DEVELOPER

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KGSNG



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EDUCATION

Mapua University (2018 - 2022)

B.S. in Manufacturing Engineering

SKILLS

BACKEND: Node.js (Proficient), Express.js (Proficient), NestJS (Proficient), Fastify (Proficient), Sequelize (Proficient), SQL (Proficient), NoSQL (MongoDB) (Proficient), Spring Boot (Proficient), Quarkus (Proficient), TypeScript (Proficient)

FRONTEND: ReactJS (Proficient), React Redux (Proficient), React-router (Proficient), Angular (Proficient), RxJS (Proficient), Vite.js (Familiar), Bulma (Familiar), Tailwind CSS (Proficient), GraphQL (Experienced), Axios (Proficient), HTML5 (Proficient), NPM (Proficient), Yarn (Proficient), Storybook (Proficient), Next.js (Proficient)

AI & Security Automation: Agentic AI Architecture, Custom AI Agents, LLM Optimization, Anti-Bloating Patterns, AI Automated Security Remediation, DevSecOps Automation, AI Vulnerability Patching Automation

SOFTSKILL: Team Collaboration, Adaptability, Problem Solving, Open-Minded, Confidence, Accountability, Communication, Time Management

CLOUD TECHNOLOGY:

- **Azure** (Proficient) — Azure Functions, Virtual Machines, Blob Storage, Azure Resource Manager (ARM), Azure Kubernetes Service (AKS), Azure CLI/PowerShell, Azure API Management, Azure Monitor, Application Insights, Azure Active Directory, Azure Key Vault, Azure SQL Database, Cosmos DB, Azure CDN
- **AWS** (Proficient) — Lambda, EC2, S3, CloudFormation, EKS, CDK, SDK, SAM, API Gateway, CloudWatch, X-Ray, Cognito, Parameter Store, Elasticsearch, RDS, DynamoDB, CloudFront, DevOps Pipeline, CI/CD

DevOps: DevOps Pipeline, CI/CD (AWS CodePipeline, GitHub Actions, GitLab CI, Azure DevOps), Environment Management, Docker, Podman, Trivy, SonarQube, Rancher, Grafana, Prometheus

Security: SSO integration (Azure AD, AWS Cognito, PingFed), OAuth 2.0, JWT, Hashing, Security Vulnerability Management, Encryption, 2FA

Tools & Technology: Git, GitLab, Bitbucket, Jira, Agile (Scrum/Kanban), Google Workspace, Microsoft Office

PROFILE

Senior Full Stack & Cloud-Native Engineer with 4+ years of experience building scalable web applications, modernizing enterprise systems, and architecting production-grade **AI/agentic engineering platforms**. Expert in transitioning **legacy architectures** to modern stacks. Spring Boot, Quarkus, Node.js, and React and in leveraging **Microsoft Azure and AWS for cloud-native, event-driven solutions**. Proven track record designing multi-account layered architectures, DevSecOps automation, and custom AI-agent systems that enforce security and code quality at scale. Seeking a senior/lead engineering role to apply this experience to a forward-thinking team.

WORK EXPERIENCE

SENIOR SOFTWARE ENGINEER

Australian Finance Group

March 2026 - Present

Project: AFG Securities Integration & Agentic Engineering Platform

- **Core Banking Data Pipeline:** Engineered an automated, daily-synchronized extraction and normalization pipeline against a legacy Core Banking System, transforming raw relational structures into a queryable, high-performing data layer backed by Amazon Aurora Serverless v2 (PostgreSQL 17) with Drizzle ORM for type-safe, zero-drift schema management.
- **Data Correlation Engine:** Built a complex aggregation workflow that joins and normalizes disparate entities across dynamic relational hierarchies, producing clean unified data assets consumed by internal applications and major third-party platforms including Zendesk, Flex, and Ray White.
- **Domain Orchestration Layer:** Designed a strict layered service architecture (Public → Domain → Supporting) across four independently deployed AWS accounts, enforcing unidirectional call chains and decoupling public API exposure from core banking data access with zero cross-layer coupling. Governed by 9 ratified Architecture Decision Records (ADRs) covering CBS routing, EventBridge technology selection, read-only CBS access, and supporting-layer publish rights.
- **Resilient Event-Driven Architecture:** Designed an asynchronous, EventBridge-based propagation system with per-entity SQS queues, multi-stage Dead Letter Queues, automated redrive, schema-validated subscribers, and archive/replay — ensuring high fault tolerance and strict data integrity across the T+1 sync cycle. Extended with a customer-info propagator stack for real-time downstream fan-out.
- **Serverless Infrastructure:** Leveraged AWS Lambda, Amazon Aurora Serverless v2, AWS API Gateway, WAF, SQS, EventBridge, and SSM Parameter Store — fully provisioned via AWS CDK (TypeScript) with multi-account deployment pipelines across sandbox, dev, SIT, UAT, and production environments.
- **Agentic AI Engineering Platform:** Architected a custom multi-agent AI system across 4 service repositories comprising 85 specialized agent configurations — including cross-service planner, per-repo planners, code reviewers, bloat checkers, SOLID reviewers and implementers, infra-stack agents, test writers, bug hunters, cost analysts, incident responders, business analysts, PR preparers, merge-conflict resolvers, and a shared adversarial design challenger (rubber-duck). Backed by 26+ reusable skill pipelines covering spec-driven development, TDD, incremental implementation, multi-agent review, doubt-driven development, observability, security hardening, and more. Enforces architectural constraints, anti-bloat patterns, SOLID principles, and multi-round code review automatically on every PR via cross-repo universal checks and mandatory design debates before every ticket.
- **Automated Security & DevSecOps:** Engineered a proprietary AI-driven security pipeline that autonomously scans repositories on a weekly scheduled cadence, triages vulnerabilities by severity and exploitability, and generates isolated, verified Pull Requests with auto-remediation patches — operating similarly to Snyk and Dependabot but with full architectural context. Covers both direct and transitive dependency vulnerabilities, with allowlist governance for unfixable bundled deps.

SENIOR ASSISTANT MANAGER - SENIOR FULLSTACK ENGINEER

East West Bank Corp.

November 2025 - March 2026

Projects: CORE, RETAIL/SALES and CORP

- **Cloud-Native & Modernization:** Engineered cloud-native applications leveraging Microsoft Azure; led a Proof of Concept (POC) for migrating services to Quarkus to optimize container startup times and reduce memory footprint.
- **API Modernization & Integration:** Architected a modernization layer to consume 3rd party SOAP web services and expose them as RESTful APIs, reducing integration complexity and enabling modern frontend consumption.
- **DevSecOps & Pipeline Automation:** Integrated Trivy into CI/CD pipelines to automate container image scanning and vulnerability detection. Developed reusable pipeline boilerplates, standardizing the deployment process across teams and reducing setup time for new microservices.
- **Security & Quality Assurance:** Eliminated 100% of project vulnerabilities identified by SonarQube, maintaining a clean security profile. Achieved a 96% unit test coverage score (JaCoCo) by implementing rigorous testing frameworks for core financial logic.
- **Database & API Design:** Architected the Fund Transfer database schema and logic, ensuring a robust, scalable flow for high-concurrency financial transactions. Created automated documentation and governance for all API contracts, ensuring consistency across the project lifecycle.

WORK EXPERIENCE

SENIOR SOFTWARE ENGINEER

Stratpoint Technologies Inc.

November 2024 - October 2025

- **Led a team of 5 software engineers** using Agile methodologies to successfully deliver a full-cycle procurement system 2 weeks ahead of schedule. Fostered collaboration with Business Analysts, QA teams, and stakeholders to align technical solutions with business requirements.
- **Engineered a scalable, RESTful backend API** using Node.js and Express, handling critical modules for Requisition Slips, Purchase Orders, Delivery Receipts, and Payment Requests. The system managed a complex data model with over 50 unique endpoints.
- **Designed and implemented a complex, configurable approval workflow** with a custom rule engine, supporting dynamic, multi-level approval chains based on departmental and financial associations. The system allowed for parallel and sequential approval logic.
- **Engineered a secure, Role-Based Access Control (RBAC) system** with comprehensive authentication features including password resets, temporary passwords, and OTP-based 2FA, securing over 100 unique permissions across different user roles.
- **Developed a reusable form component library leveraging React**, react-hook-form, and zod for schema-based validation, which reduced development time by 30% and served as the foundation for the application's UI.
- **Created advanced, accessible UI components** including a feature-rich Select dropdown with server-side searching, debouncing, and infinite scrolling for paginated data, and a resilient file upload system with real-time progress tracking.
- **Built robust data synchronization services** to ensure data integrity by periodically syncing critical business data (e.g., Suppliers, Companies, Items) from external legacy APIs.
- **Implemented a comprehensive, event-driven notification system** to provide real-time updates to users via email and in-app alerts on status changes, approval requests, and background task completion.
- **Engineered a dynamic report generation service** to create complex, multi-page PDF and Excel reports programmatically, handling sophisticated logic for pagination and chunking large datasets to ensure professional and accurate report layouts.

SOFTWARE ENGINEER

ASURION

March 2023 - Oct 2024

Customer Warranty Tracker(Front-office Application)

- Over 6 months, developed a cost-optimized, serverless application that significantly reduced operational overhead and infrastructure costs by minimizing EC2 and RDS usage.
- This solution was built with Go (for backend logic) and React Native (for the mobile front-end), and it leveraged AWS Lambda to automate CI/CD pipelines and environment management.

Forecasting, Analytics, Warehousing Application (Back-office)

- Developed and maintained robust, scalable, and highly available **Node.js** and **React** applications using an Agile methodology.
- Successfully migrated a legacy ES4 application to a **Node.js-based** NestJS framework (Typescript base framework), which improved code maintainability and reduced load times by 20%.
- Implemented a high-performance microservices architecture using Node.js and leveraged **AWS services like EC2, S3, Lambda, and RDS**.
- Built a reusable UI component library with **Storybook** for **React**, accelerating development and ensuring design consistency across projects.
- Optimized application performance by using AWS tools such as X-Ray for tracing and CloudWatch for monitoring, and fine-tuned **Node.js** application performance.
- Proactively addressed security vulnerabilities by implementing best practices, including data validation, encryption, and session caching.
- Integrated **PingFed SSO** for enhanced security and streamlined user access.
- Collaborated effectively with cross-functional teams to deliver high-quality solutions.

SOFTWARE ENGINEER INTERN

ASURION

November 2022 - February 2023

- **Serverless Application:** Maintained a complex **serverless** application built on **AWS Lambda**, proactively identifying and resolving performance **bottlenecks** and **security vulnerabilities**, ensuring smooth operation and data integrity.
- **Full-Stack Development:** Actively participated in real-world projects, contributing to both frontend and backend development.
- **Frontend Development:** Developed comprehensive features using ReactJS, leveraging Redux for efficient state management.
- **Backend Integration:** Collaborated with backend teams to understand API endpoints and seamlessly integrate frontend components with backend services.
- **Agile Development:** Adhered to Agile methodologies (Scrum, Kanban) to deliver high-quality features on time and within budget.
- **Continuous Improvement:** Stayed up-to-date with the latest frontend technologies and industry best practices to enhance code quality and user experience.